OCTO-EARTH

Calculation report: mission, the three enablers and every instrument

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All numbers are model results of a physically hypothetical scenario, not engineering designs.

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1. What the system is and the three enablers

OCTO-EARTH moves Earth (with its Moon) from its orbit at 1 AU to a circular orbit at **2.7 AU** within **10,000 simulated years** by throwing mass away from Earth (momentum exchange: Earth recoils the other way). The thrown mass comes from an asteroid docked to Earth, the energy comes from sunlight collected near the Sun and beamed to Earth as microwaves, and Earth's climate is kept livable far from the Sun by mirrors that reflect extra sunlight onto it. A reinforcement-learning agent (PPO) chooses every 100 years how hard, how fast and in which direction to throw, and how much mirror light to send to Earth.

Exactly three enablers. An enabler is a technology the model ASSUMES can be built without showing it. The model allows only three; each one is a complete system made of the instruments listed under it. Everything else in the model is ordinary physics (gravity, momentum, energy, radiation) or ordinary control software.

	Enabler	What it does in the model	Instruments
E1	Earth launcher system: docked asteroid + launch ring with 8 arms + arm radiators	Throws packets of up to 1e18 kg of asteroid material at up to 5,000 km/s, 8 times per year (all 8 arms throw one packet together), with 50 % of the drawn energy ending up as packet kinetic energy; the arms reject their waste heat away from Earth; the launches do not damage Earth.	• Asteroid reservoir (docked to Earth) • Launch ring + 8 launch arms (mass driver) • Arm power conditioning + energy reserve • Directional arm radiators
E2	Solar power belt with microwave power beaming (collectors + transmitters + arm rectennas)	5 million thin-film collectors at 0.5 AU intercept 30 % of the Sun's output, turn it into electricity and beam it as 5.8 GHz microwaves (30 km phased arrays) to 1,000 km rectennas on the 8 arms; they hold their slots by sail tilt, keep a clear sunlight lane for Earth and Mars, keep beam spill off Earth and are built up / replaced by factories using Mercury material.	• Thin-film solar collector (x 5 million) • Microwave phased-array transmitter (one per collector) • Rectenna on each of the 8 arm tips • Sail-tilt station keeping (on each collector) • Clear sunlight lane • Collector factories (Mercury material)
E3	Space mirror ring around Earth (soletta)	1.6e15 m^2 of aluminium-film mirrors (8e12 kg) in a terminator orbit 20 Earth radii out reflect extra sunlight onto Earth: up to 1.10 x today's sunlight at 2.7 AU. The agent sets how much (mirrors not needed are turned so their light passes Earth).	• Mirror film panels (aluminium, 5 g/m^2) • Terminator orbit at 20 Earth radii + pointing (45 degree tilt)

Chain of cause and effect

Sunlight → **E2** collectors (electricity) → **E2** transmitters (microwaves) → **E2** rectennas on the arms (electricity) → **E1** launch arms throw asteroid packets (**E1** reservoir) → Earth recoils, its orbit grows → less sunlight at Earth → **E3** mirrors add sunlight → climate stays near 288 K. Waste heat of E1 and E2 is sent away from Earth by the E1 radiators and the E2 spill control; the small share that still reaches Earth is part of the climate calculation.

Key numbers at a glance

Quantity	Value
Required speed change of Earth (slow spiral)	11,658 m/s

Orbital energy Earth+Moon must gain	1.70×10^{33} J
Minimum launch energy (whole asteroid, slowest launch)	1.66×10^{35} J
Asteroid (reaction mass)	3.00×10^{22} kg = 0.408 Moon masses
Launch speed used by the hand-designed strategy	2,480 km/s = 0.83 % of light speed
Sunlight intercepted by the collector belt	1.15×10^{26} W (30 % of the Sun)
Electricity at the arms (Earth at 1 AU / at the target)	1.11×10^{25} W (weakest point)
Space-mirror capacity at the target	1.10 × today's sunlight on Earth
Analytic verdict	**FEASIBLE within this model (physically hypothetical scale)**
Tested mission (hand-designed, high fidelity)	success: a = 2.6903 AU, e = 0.0072 after 3,600 y; surface 288.4 K

Numbers are labelled **C1, C2, …**. Each shows the formula, the values put in, the result and the function in `octoearth_rl.py` that does the same calculation in the simulation. Values marked *indicative* are NOT used by the simulation (they only help to judge an instrument).

2. Physical constants and mission inputs

Constant	Value	Unit
Gravitational constant G	6.67430×10^{-11}	m^3 kg^-1 s^-2
Sun mass	1.98841×10^{30}	kg
Earth mass	5.97217×10^{24}	kg
Moon mass	7.34579×10^{22}	kg
Mars mass	6.41691×10^{23}	kg
Mercury mass	3.30100×10^{23}	kg
Venus mass	4.86731×10^{24}	kg
Jupiter mass	1.89852×10^{27}	kg
Mars radius	3.38950×10^{6}	m
Astronomical unit (AU)	1.49598×10^{11}	m
Year	3.15576×10^{7}	s
Earth radius	6.37100×10^{6}	m
Arm length	1.27420×10^{7}	m
Launch-ring radius	7.96375×10^{6}	m
Grab point height on the ring axis	7.96375×10^{6}	m
Moon distance	3.84400×10^{8}	m
Solar constant at 1 AU	1,361.00	W/m^2
Speed of light	2.99792×10^{8}	m/s

Mission input	Value
Target orbit radius a_t	2.7 AU (tolerance ±0.05 AU, e ≤ 0.05, i ≤ 5.0°)
Mission time limit T	10,000 years = 3.16×10^{11} s
Decision interval	100 years (100 decisions)
Climate target	288 K ± 5 K

The full list of all parameters with their values, units and meaning is in the appendix.

3. Mission requirement: how much push, energy and time

C1 **Mass that has to be moved**

Formula $m_{sys} = M_{Earth} + m_{asteroid} + M_{Moon}$

Values $5.9722\times10^{24} + 3.00\times10^{22} + 7.346\times10^{22}$ kg

Result **6.0756×10^{24} kg**

code: feasibility()

C2 **Gravitational parameter of the Sun-Earth system**

Formula $\mu = G\cdot M_{Sun} + G\cdot m_{sys}$

Values $1.32712\times10^{20} + 4.055\times10^{14}$ m³/s²

Result **1.32713×10^{20} m³/s²**

code: feasibility()

C3 **Circular orbital speed today (1 AU)**

Formula $v_0 = \sqrt{(\mu / a_0)}$

Values $\sqrt{(1.327\times10^{20} / 1.4960\times10^{11})}$

Result **29,784.7 m/s**

code: feasibility()

C4 **Circular orbital speed at the target (2.7 AU)**

Formula $v_t = \sqrt{(\mu / a_t)}$

Values $\sqrt{(1.327\times10^{20} / 4.0391\times10^{11})}$

Result **18,126.4 m/s**

code: feasibility()

C5 **Required speed change (slow outward spiral)**

Formula $\Delta v = v_0 - v_t$

Values 29,784.7 − 18,126.4

Result **11,658 m/s**

Meaning A slow spiral (small pushes along the motion over thousands of years) needs exactly the difference of the two circular speeds.

code: feasibility() dv_spiral

C6 **Reference: Hohmann transfer (two short burns)**

Formula $a_{tr} = (a_0+a_t)/2$; $\Delta v = |\sqrt{(\mu(2/a_0-1/a_{tr}))} - v_0| + |v_t - \sqrt{(\mu(2/a_t-1/a_{tr}))}|$

Values a_{tr} = 1.850 AU

Result **10,997 m/s**

Meaning Cheaper in speed change, but would need two enormous short pushes; the launcher can only push gradually, so the spiral value is used.

code: feasibility() dv_hoh

C7 **Orbital energy Earth + Moon (+asteroid) must gain**

Formula $\Delta E = G\, M_{Sun}\, m_{sys}\, (1/(2a_0) - 1/(2a_t))$

Values $1.3271\times10^{20} \times 6.076\times10^{24} \times (1/2.992\times10^{11} - 1/8.078\times10^{11})$

Result **1.70×10^{33} J**

code: feasibility() dE_orb

C8 Orbital angular momentum change

Formula $\Delta L = m_{sys} (\sqrt{(\mu\ a_t)} - \sqrt{(\mu\ a_0)})$

Result **1.74×10^{40} kg m²/s**

code: feasibility() dL_orb

C9 Average power just for the orbital energy

Formula $P = \Delta E / T$

Values 1.70×10^{33} J / 3.16×10^{11} s

Result **5.38×10^{21} W = 30,982 × all sunlight Earth intercepts**

code: feasibility()

C10 Escape speed from Earth + docked asteroid

Formula $v_{esc} = \sqrt{(2\ G\ (M_{Earth}+m_{ast}) / R_{Earth})}$

Values $\sqrt{(2 \times 6.6743\times10^{-11} \times 6.002\times10^{24} / 6.371\times10^{6})}$

Result **11,214 m/s**

Meaning A packet launched slower than this falls back; faster packets keep $v_\infty = \sqrt{(u^2 - v_{esc}^2)}$.

code: feasibility() vesc / launch_vinf()

C11 Rocket equation: mass ratio if the whole asteroid is thrown

Formula $MR = m_{sys} / (m_{sys} - m_{ast})$

Values $6.0756\times10^{24} / 6.0456\times10^{24}$

Result **1.004962 (ln MR = 0.004950)**

code: feasibility()

C12 Speed change possible at the maximum launch speed

Formula $\Delta v_{cap} = v_{\infty,max} \cdot \ln MR$, $v_{\infty,max} = \sqrt{(u_{max}^2 - v_{esc}^2)}$

Values $5.0000\times10^{6} \times 0.004950$

Result **24,750 m/s (needed 11,658 m/s)**

code: feasibility() dv_cap_mass

C13 Reaction mass needed at the maximum launch speed

Formula $m = m_{sys} (1 - e^{-\Delta v / v_{\infty,max}})$

Result **1.41×10^{22} kg = 0.193 Moon masses**

code: feasibility() mass_needed_umax

C14 Slowest launch speed that works with the whole asteroid (minimum energy)

Formula $v_\infty = \Delta v / \ln MR$; $u = \sqrt{(v_\infty^2 + v_{esc}^2)}$

Values v_∞ = 11,658 / 0.004950 = 2.3552×10^{6} m/s

Result **u = 2.3552×10^{6} m/s**

code: feasibility() u_req

C15 Minimum launch energy

Formula $E_{min} = ½\ m_{ast}\ u^2 / \eta_{launch}$

Values $0.5 \times 3.00\times10^{22} \times (2.355\times10^{6})^2 / 0.5$

Result **1.66×10^{35} J (average 5.27×10^{23} W over the mission)**

code: feasibility() e_draw_min

C16 Shortest possible transfer time (maximum packet rate and speed)

Formula $\Delta v \text{ per year} = n \cdot m_{pk} \cdot v_{\infty,max} / m_{sys}$; $t_{min} = \Delta v / (\Delta v \text{ per year})$

Values $8 \times 1.00\times10^{18} \times 5.000\times10^{6} / 6.076\times10^{24} = 6.584$ m/s per year

Result **1,771 years**

code: feasibility() years_min

C17 Mass + time limits together

Formula $v_\infty = \max(\Delta v/\ln MR, \Delta v\, m_{sys}/(T\, n\, m_{pk}))$; $m = m_{sys}(1-e^{-\Delta v/v_\infty})$; $E = \tfrac{1}{2}\, m\, u^2/\eta$

Result **u = 2.3552×10^6 m/s, m = 3.00×10^22 kg, E = 1.66×10^35 J**

code: feasibility() combined

C18 Launcher peak power (maximum packet rate and speed)

Formula $P_{peak} = n \cdot m_{pk} \cdot \tfrac{1}{2}\, u_{max}^2 / \eta / \text{year}$

Values $8 \times 1.00\times10^{18} \times 0.5 \times (5.00\times10^{6})^2 / 0.5 / 3.1558\times10^{7}$

Result **6.34×10^24 W**

code: feasibility() p_peak

C19 Energy available over the mission

Formula $E_{avail} = \text{reserve} + P_{elec,min} \cdot T$

Values $3.00\times10^{35} + 1.11\times10^{25} \times 3.16\times10^{11}$

Result **3.82×10^36 J**

code: feasibility() e_avail

Check	Result	Detail
mass	OK	needs launch speed 2.355e+06 m/s with ALL 3.00e+22 kg; max allowed 5.000e+06 m/s (mass-limited delta-v capacity 2.475e+04 m/s vs required 1.166e+04 m/s)
energy	OK	minimum energy drawn 1.664e+35 J vs available 3.817e+36 J (reserve 3.000e+35 J + collectors 1.114e+25 W x 10,000 y)
rate	OK	at max packet rate and speed the transfer needs >= 1,771 years (mission 10,000 y)
combined	OK	slowest speed meeting mass AND time limits 2.355e+06 m/s -> needs 3.000e+22 kg and 1.664e+35 J

Verdict: FEASIBLE within this model (physically hypothetical scale). The realistic preset (Ceres-mass asteroid, 30 km/s launcher, Earth's own sunlight as energy) is infeasible: verdict *INFEASIBLE with these parameters (failed: mass, energy, rate, combined; mass-limited delta-v reaches 3.71e-04 of the requirement)*.

4. Enabler E1 - Earth launcher system

Earth launcher system: docked asteroid + launch ring with 8 arms + arm radiators
Throws packets of up to 1e18 kg of asteroid material at up to 5,000 km/s, 8 times per year (all 8 arms throw one packet together), with 50 % of the drawn energy ending up as packet kinetic energy; the arms reject their waste heat away from Earth; the launches do not damage Earth.

Gap to reality: No known structure or material; each launch carries ~1e30 J (millions of Chicxulub impacts); moving a 3e22 kg body to Earth and docking it is far beyond any technology; ~1e19 m^2 of 1,000 K radiator per mission.

Modelled: Momentum, energy, escape from Earth (+Moon) gravity, packet fates, finite reservoir, waste heat into the climate model. NOT modelled: the hardware, launch shock, atmosphere loss and heating of Earth (problem P6).

Instrument E1.1 - Asteroid reservoir (docked to Earth)

Use: supplies the reaction mass. Earth can only change its speed by throwing mass the other way (conservation of momentum); the reservoir is finite and is used up packet by packet.

C20 Reservoir size compared with the Moon and Earth

Formula	m_{ast} / M_{Moon}, m_{ast} / M_{Earth}
Values	$3.00\times10^{22} / 7.346\times10^{22}$; $3.00\times10^{22} / 5.972\times10^{24}$
Result	**0.408 Moon masses; 0.502 % of Earth**

code: Config.asteroid_mass_kg

C21 Surface gravity with the asteroid attached (indicative)

Formula	$g = G\,(M_{Earth} + m_{ast}) / R_{Earth}^2$
Result	**9.8696 m/s² (today 9.8203 m/s², +0.50 %)**
Meaning	The model treats Earth + asteroid as one body (its gravity counts in every orbit calculation).

code: indicative

C22 Number of full packets in the reservoir

Formula	$N = m_{ast} / m_{pk,max}$
Values	$3.00\times10^{22} / 1.00\times10^{18}$
Result	**30,000 packets = 3,750 years at the maximum rate**

C23 Tested mission: reservoir used

Formula	m_{used} / m_{ast}
Values	$2.86\times10^{22} / 3.00\times10^{22}$
Result	**95.2 % in 28,560 packets**

code: summary() mass_used_frac

Instrument E1.2 - Launch ring with 8 arms (mass driver)

Use: accelerates each packet to the commanded speed u (between 20.0 and 5,000 km/s) in the commanded direction; 8 launches per year. All 8 arms throw ONE packet together (group throw), so each arm carries 1/8 of the packet and of the energy.

C24 **Launch speed chosen by the hand-designed strategy**

Formula $v_\infty = \max(\Delta v / \ln(m_{sys}/(m_{sys}-0.95\ m_{ast})),\ \Delta v\ m_{sys} / (0.85\ T\ n\ m_{pk}));\ u = \sqrt{(v_\infty^2+v_{esc}^2)}$

Values $\max(2.479\times10^{6},\ 1.042\times10^{6})$ m/s

Result **$u = 2.4795\times10^{6}$ m/s = 2,480 km/s (0.83 % of c)**

Meaning Lowest speed that uses 95 % of the asteroid and still finishes within 85 % of the mission time.

code: heuristic_speed()

One full packet at the strategy speed (u = 2,480 km/s)

C25 **Kinetic energy given to packet + Earth (two-body, reduced mass)**

Formula $\mu_{red} = m_{pk}(M-m_{pk})/M;\ E_{kin} = \tfrac{1}{2}\ \mu_{red}\ u^2$

Values $\mu_{red} = 1.00000\times10^{18}$ kg; $0.5 \times 1.000\times10^{18} \times (2.480\times10^{6})^2$

Result **3.07×10^{30} J**

code: eject_core()

C26 **Energy drawn from the power supply**

Formula $E_{draw} = E_{kin} / \eta_{launch}$

Values $3.07\times10^{30} / 0.5$

Result **6.15×10^{30} J per packet, 7.68×10^{29} J per arm; 6.15×10^{6} × a Chicxulub impact (~10^{24} J)**

code: eject_core()

C27 **Packet speed after leaving Earth's and the Moon's gravity**

Formula $v_\infty = \sqrt{(u^2 - 2GM/R_{Earth} - 2GM_{Moon}/d_{Moon})}$

Values $\sqrt{((2.480\times10^{6})^2 - 1.258\times10^{8} - 25{,}509)}$

Result **2.47949×10^{6} m/s**

code: eject_core() (fast model)

C28 **Recoil: speed change of Earth + Moon per packet**

Formula $\Delta v_E = m_{pk}\ v_\infty / (M_{Earth}+m_{ast}+M_{Moon})$

Values $1.00\times10^{18} \times 2.479\times10^{6} / 6.0756\times10^{24}$

Result **0.408 m/s (impulse 2.48×10^{24} N s)**

code: _fast_eject() / ledger LG_IMPULSE_TOTAL

C29 **Full-throttle push per year and per decision**

Formula $\Delta v_{yr} = n \cdot \Delta v_E$

Values 8×0.408

Result **3.26 m/s per year, 326 m/s per 100-year decision; Δv needs 3,571 years of full thrust**

code: policy_heuristic() dv_max

C30 **Average launcher power at full throttle**

Formula $P = n \cdot E_{draw} / year$

Values $8 \times 6.15\times10^{30} / 3.1558\times10^{7}$

Result **1.56×10^{24} W (1.95×10^{23} W per arm)**

One full packet at the maximum speed (u = 5,000 km/s)

C31 Kinetic energy given to packet + Earth (two-body, reduced mass)

Formula $\mu_{red} = m_{pk}(M-m_{pk})/M$; $E_{kin} = \tfrac{1}{2}\,\mu_{red}\,u^2$

Values $\mu_{red} = 1.00000\times10^{18}$ kg; $0.5 \times 1.000\times10^{18} \times (5.000\times10^{6})^2$

Result **1.25×10^{31} J**

code: eject_core()

C32 Energy drawn from the power supply

Formula $E_{draw} = E_{kin} / \eta_{launch}$

Values $1.25\times10^{31} / 0.5$

Result **2.50×10^{31} J per packet, 3.12×10^{30} J per arm; 2.50×10^{7} × a Chicxulub impact (~10^{24} J)**

code: eject_core()

C33 Packet speed after leaving Earth's and the Moon's gravity

Formula $v_\infty = \sqrt{(u^2 - 2GM/R_{Earth} - 2GM_{Moon}/d_{Moon})}$

Values $\sqrt{((5.000\times10^{6})^2 - 1.258\times10^{8} - 25{,}509)}$

Result **4.99999×10^{6} m/s**

code: eject_core() (fast model)

C34 Recoil: speed change of Earth + Moon per packet

Formula $\Delta v_E = m_{pk}\, v_\infty / (M_{Earth}+m_{ast}+M_{Moon})$

Values $1.00\times10^{18} \times 5.000\times10^{6} / 6.0756\times10^{24}$

Result **0.823 m/s (impulse 5.00×10^{24} N s)**

code: _fast_eject() / ledger LG_IMPULSE_TOTAL

C35 Full-throttle push per year and per decision

Formula $\Delta v_{yr} = n \cdot \Delta v_E$

Values 8×0.823

Result **6.58 m/s per year, 658 m/s per 100-year decision; Δv needs 1,771 years of full thrust**

code: policy_heuristic() dv_max

C36 Average launcher power at full throttle

Formula $P = n \cdot E_{draw}$ / year

Values $8 \times 2.50\times10^{31} / 3.1558\times10^{7}$

Result **6.34×10^{24} W (7.92×10^{23} W per arm)**

C37 Time between launches

Formula $\Delta t = 1$ year / n

Values 365.25 d / 8

Result **45.66 days**

code: fast_interval()

C38 Mass per arm in a group throw

Formula $m_{arm} = m_{pk} / 8$

Values $1.00\times10^{18} / 8$

Result **1.25×10^{17} kg**

code: N_ARMS = 8

C39 **Tested mission: total impulse and speed change delivered**

Formula $\Delta v = J_{total} / M_{Earth+Moon+ast}$

Values 7.08×10^{28} N s / 6.0756×10^{24} kg

Result **11,655 m/s (required by the spiral estimate: 11,658 m/s)**

Meaning Compare with the slow-spiral estimate C5: the hand-designed strategy pushes almost purely along Earth's motion, so the delivered speed change is close to the ideal value.

code: ledger LG_IMPULSE_TOTAL

C40 **Tested mission: packet kinetic energy and launcher draw**

Formula $\eta = E_{kin} / E_{draw}$

Values 8.78×10^{34} / 1.76×10^{35}

Result **0.5000 (= launch_efficiency 0.5)**

code: ledger LG_LAUNCH_KE, LG_E_DRAWN

C41 **Tested mission: average launcher power**

Formula $P = E_{draw} / t$

Values 1.76×10^{35} / (3,600 y × 3.1558×10^{7} s)

Result **1.55×10^{24} W**

Launcher geometry: asteroid on the ring axis, arms 2 Earth radii long (update p36)

The 8 arm roots sit on a ring around Earth (plane perpendicular to Earth's orbit normal). The docked asteroid sits on the ring's **axis**; its Earth-facing side rests on the grab point, so every arm root is exactly the same distance from it and all 8 arms reach it equally. The docking clamps lower the asteroid as it shrinks, so the grab point never moves. Each arm is 2 Earth radii long.

C42 **Distance from every arm root to the grab point**

Formula $d = \sqrt{(\rho_{ring}^2 + z_{grab}^2)} \le L_{arm}$

Values $\sqrt{(1.25^2 + 1.25^2)}$ R_E; L_{arm} = 2.0 R_E = 12,742 km

Result **1.7678 R_E = 11,262 km for all 8 roots (spread 0e+00); reach to spare 1,480 km**

code: launcher_geometry()

C43 **Asteroid size and position**

Formula $r_{ast} = (3\, m_{ast} / (4\pi\rho))^{1/3}$; $z_{centre} = z_{grab} + r_{ast}$

Values ρ = 2000 kg/m³ (assumed), $m_{ast} = 3.00\times10^{22}$ kg

Result **r_{ast} = 1,530 km; centre 1.490 R_E from Earth's centre, 1.945 R_E from every root (≤ arm length); 1,593 km above the surface**

code: launcher_geometry()

C44 **Stroke of the joint throw (all 8 arms keep hold)**

Formula largest s with $|G + s\cdot e - root_k| \le L_{arm}$ for all k (worst direction e in the ring plane)

Values bisection over directions

Result **s = 1,983 km (0.311 R_E; up to 0.330 R_E between two arms); release point 1.288 R_E from Earth's centre**

code: launcher_geometry()

C45 **Acceleration the arms must give a packet over that stroke**

Formula $a = u^2 / (2\ s)$; $t = 2\ s / u$

Values u = 2,480 km/s (strategy) and 5,000 km/s (maximum); s = 1,983 km

Result **1.55×10^6 m/s² (1.58×10^5 g) for 1.60 s; at the maximum speed 6.30×10^6 m/s²**

Meaning Far beyond any material: part of enabler E1's gap.

code: launcher_geometry()

C46 **Load on the docking clamps (asteroid's weight)**

Formula $F = m_{ast}\ G\ M_{Earth} / z_{centre}^2$

Values 3.00×10^{22} × 4.423 m/s²

Result **1.33×10^{23} N**

code: launcher_geometry()

C47 **What the physics simplifies (and why it is allowed)**

Formula $\Delta E/E = (GM/R)(1 - R/r_{release}) / (u^2/2)$; COM shift = $m_{ast}\ z_{centre}/(M+m_{ast})$

Values escape energy from Earth's surface instead of the release point; asteroid merged with Earth

Result **surface launch needs $+4.57 \times 10^{-6}$ more energy (conservative); centre of mass off by 47 km; tidal error on the Moon 9.41×10^{-6} of Earth's pull**

code: launcher_geometry(), check 21

Not modelled (assumed by E1): the arms' structure and materials, the acceleration of the packets inside the arms, and the effect of each launch on Earth itself (shock, atmosphere loss, heating). Open problem P6.

Instrument E1.3 - Arm power conditioning and energy reserve

Use: the arms take electricity from the rectennas (E2) first; if a launch needs more than the beams deliver in that interval, the rest comes from a finite reserve. Unused beam energy is not sent (the beams are throttled to demand); nothing is created from nothing.

C48 **Reserve size compared with the minimum mission energy**

Formula reserve / E_{min}

Values 3.00×10^{35} / 1.66×10^{35}

Result **1.803**

code: Config.energy_budget_j

C49 **Energy offered by the beams in one decision interval (Earth at 1 AU)**

Formula $E_{beam} = P_{elec} \cdot \Delta t_{decision}$

Values 1.29×10^{25} W × 100 y

Result **4.07×10^{34} J**

code: _PowerSupply._with_supply()

C50 **Tested mission: where the launch energy came from**

Formula $E_{draw} = E_{beams} + E_{reserve}$

Values 1.76×10^{35} + 0

Result **1.76×10^{35} J; beams used 12.8 % of what they could offer**

code: summary() energy_from_collectors_J

Instrument E1.4 - Directional arm radiators

Use: the launcher loses (1−η) = 50 % of the energy it draws as heat, and the rectennas lose ($1-\eta_r$) = 15 % of the microwaves they catch. The radiators send this heat to space, away from Earth; only the share arm_heat_to_earth_frac = 1.00×10^{-9} reaches Earth.

C51 **Tested mission: arm waste heat**

Formula $Q = E_{\text{rectenna heat}} + E_{\text{launcher heat}}$

Values $3.10\times10^{34} + 8.78\times10^{34}$ J

Result **1.19×10^{35} J = 1.05×10^{24} W on average = 6.02×10^{6} × the sunlight Earth intercepts**

code: settle_interval()

C52 **Radiator area needed at 1,000 K (indicative, black radiator, one side)**

Formula $A = Q / (\sigma\ T\&sup4;)$

Values 1.05×10^{24} / (5.670×10^{-8} × 1000&sup4;)

Result **1.84×10^{19} m² in total, 2.30×10^{18} m² per arm = 18,074 × Earth's disc**

code: summary() radiator_area_1000K_m2

C53 **Heat that still reaches Earth**

Formula $q_{\text{Earth}} = f_{\text{arm}}\cdot Q + f_{\text{spill}}\cdot E_{\text{spill}}$

Values 1.00×10^{-9} × 1.19×10^{35} + 1.00×10^{-9} × 5.38×10^{33}

Result **1.24×10^{26} J = 1.09×10^{15} W = 0.00630 × Earth's sunlight**

code: _climate_step() q_w

Without the directional radiators a geometric leak of ~8 % would put 8.36×10^{22} W (4.82×10^{5} × Earth's sunlight) on Earth: the oceans would boil. This is why the radiators are part of enabler E1.

Mars safety limits (control software, not an enabler)

Earth's growing orbit crosses Mars's orbit. The safety shield forecasts the Earth-Mars distance and overrides the agent (coast or full thrust) when a pass would come too close. The safe distance is the largest of three limits:

C54 **Tide limit: Mars's tide on Earth ≤ 1 % of the Moon's**

Formula $d_{\text{tide}} = d_{\text{Moon}}\,(GM_{\text{Mars}} / (GM_{\text{Moon}}\cdot k_{\text{tide}}))^{1/3}$

Values 3.844×10^{8} × (4.283×10^{13} / (4.903×10^{12} × 0.01))$^{1/3}$

Result **3.67×10^{6} km**

code: mars_safety_limits()

C55 **Moon-orbit limit: Mars's pull on the Moon's orbit ≤ 0.1 % of Earth's**

Formula $d_{\text{moon}} = d_{\text{Moon}}\,(2\,GM_{\text{Mars}} / (GM_{\text{Earth}}\,k_{\text{moon}}))^{1/3}$

Values $k_{\text{moon}} = 0.001$

Result **2.30×10^{6} km**

code: mars_safety_limits()

C56 **Safe distance and shield caution distance**

Formula $d_{\text{safe}} = \max(d_{\text{tide}}, d_{\text{moon}}, d_{\text{min}})$; $d_{\text{caution}} = k_c\,d_{\text{safe}}$

Values $\max(3.67\times10^{6}, 2.30\times10^{6}, 2.00\times10^{6})$ km; $k_c = 2.0$

Result **d_{safe} = 3.67×10^{6} km = 0.0246 AU (governed by: tide on Earth); caution 0.0491 AU**

code: mars_safety_limits()

Tested mission: closest Earth-Mars pass 0.0502 AU (≥ 0.0246 AU: safe), largest tide ratio 0.00117, 272 shield overrides. Mars's own orbit changed by Δa = 0.0340 AU, Δe = -9.08×10^{-4}.

Mercury, Venus and Jupiter (update p35; Saturn and beyond not modelled)

Mercury, Venus and Jupiter start on their real (J2000) orbits and are integrated in both models: they pull on Earth, the Moon (high-fidelity model), Mars and the fragments, and Earth pulls back (two-way gravity). The same three safety rules as for Mars are applied to each planet; because the tide limits scale with GM, the safe distance scales with $GM^{1/3}$. Coming closer ends the mission (failure planet_unsafe_approach). The shield's forecast still uses only Sun + Earth + Mars.

C57 Safe distance from Mercury

Formula $d_{safe} = \max(d_{Moon}(GM/(GM_{Moon}k_{tide}))^{1/3}, d_{Moon}(2GM/(GM_{Earth}k_{moon}))^{1/3}, d_{min})$

Values GM = 2.203×10^{13} m³/s²; tide 2.94×10^{6} km, Moon-orbit 1.84×10^{6} km, minimum 2.00×10^{6} km

Result **2.94×10^{6} km = 0.0197 AU (governed by: tide on Earth)**

code: planet_safety_limits()

C58 Safe distance from Venus

Formula $d_{safe} = \max(d_{Moon}(GM/(GM_{Moon}k_{tide}))^{1/3}, d_{Moon}(2GM/(GM_{Earth}k_{moon}))^{1/3}, d_{min})$

Values GM = 3.249×10^{14} m³/s²; tide 7.22×10^{6} km, Moon-orbit 4.52×10^{6} km, minimum 2.00×10^{6} km

Result **7.22×10^{6} km = 0.0483 AU (governed by: tide on Earth)**

code: planet_safety_limits()

C59 Safe distance from Jupiter

Formula $d_{safe} = \max(d_{Moon}(GM/(GM_{Moon}k_{tide}))^{1/3}, d_{Moon}(2GM/(GM_{Earth}k_{moon}))^{1/3}, d_{min})$

Values GM = 1.267×10^{17} m³/s²; tide 5.28×10^{7} km, Moon-orbit 3.31×10^{7} km, minimum 2.00×10^{6} km

Result **5.28×10^{7} km = 0.3526 AU (governed by: tide on Earth)**

code: planet_safety_limits()

C60 Pull of Mercury on Earth (two-way gravity in both models)

Formula $a = GM / d^2$ (and Earth pulls back with $G\,M_{Earth} / d^2$)

Values GM = 2.203×10^{13} m³/s²

Result **at the closest approach of the tested mission (0.550 AU): 3.25×10^{-9} m/s²; at 2.7 AU with Mercury at its nearest (2.31 AU): 1.84×10^{-10} m/s² = 2.26×10^{-7} × the Sun's pull there**

code: _planet_kicks() (fast) / nb_accel() (N-body)

C61 Pull of Venus on Earth (two-way gravity in both models)

Formula $a = GM / d^2$ (and Earth pulls back with $G\,M_{Earth} / d^2$)

Values GM = 3.249×10^{14} m³/s²

Result **at the closest approach of the tested mission (0.267 AU): 2.03×10^{-7} m/s²; at 2.7 AU with Venus at its nearest (1.98 AU): 3.71×10^{-9} m/s² = 4.57×10^{-6} × the Sun's pull there**

code: _planet_kicks() (fast) / nb_accel() (N-body)

C62 **Pull of Jupiter on Earth (two-way gravity in both models)**

Formula $a = GM / d^2$ (and Earth pulls back with $G\,M_{Earth} / d^2$)

Values $GM = 1.267 \times 10^{17}$ m³/s²

Result **at the closest approach of the tested mission (2.325 AU): 1.05×10^{-6} m/s²; at 2.7 AU with Jupiter at its nearest (2.50 AU): 9.04×10^{-7} m/s² = 0.00111 × the Sun's pull there**

code: _planet_kicks() (fast) / nb_accel() (N-body)

planet	closest approach (tested mission, N-body)	safe distance	verdict	orbit change Δa
Mercury	0.550 AU (year 0)	0.0197 AU	safe	+2.1e-06 AU
Venus	0.267 AU (year 8)	0.0483 AU	safe	-9.3e-06 AU
Jupiter	2.325 AU (year 3567)	0.3526 AU	safe	+1.5e-04 AU

Jupiter, 318 Earth masses, dominates the effect on Earth's final orbit; the 8:3 mean-motion resonance with Jupiter lies at about 2.705 AU, very close to the 2.7 AU target, so near the target Jupiter's periodic pull slowly changes Earth's eccentricity. Over the simulated mission this stays within the tolerance; over millions of years it is NOT checked here (long-term stability would need a dedicated study).

5. Enabler E2 - Solar power belt with microwave power beaming

Solar power belt with microwave power beaming (collectors + transmitters + arm rectennas)
5 million thin-film collectors at 0.5 AU intercept 30 % of the Sun's output, turn it into electricity and beam it as 5.8 GHz microwaves (30 km phased arrays) to 1,000 km rectennas on the 8 arms; they hold their slots by sail tilt, keep a clear sunlight lane for Earth and Mars, keep beam spill off Earth and are built up / replaced by factories using Mercury material.

Gap to reality: A Dyson-swarm-scale project (~1e13 x all solar panels ever built); record power beaming is ~kW over km, here ~1e24 W over 0.5-3 AU with pointing to ~1e-12 rad; beam intensity at the rectennas ~2e8 x sunlight.

Modelled: Panel temperature and efficiency, degradation and replacement, build-up, manufacturing energy and mass, Goubau diffraction capture, Sun blocking, every loss in the energy ledger, clear lanes, the orbits of 12+12 followed collectors with / without station keeping (beam recoil, planets, light pressure).

Instrument E2.1 - Thin-film solar collectors

Use: intercept sunlight and turn it into electricity. The belt is a band around the Sun at 0.5 AU that covers the fraction f = 0.3 of a full sphere, so it intercepts exactly that fraction of all sunlight.

C63 Belt area

Formula $A = f \cdot 4\pi R^2$

Values $0.3 \times 4\pi \times (7.480 \times 10^{10})^2$

Result **2.11×10^{22} m² = 4.14×10^{7} × Earth's surface**

code: collector_design()

C64 Band half-width (latitude seen from the Sun)

Formula $\beta_{band} = \arcsin f$

Values $\arcsin 0.3$

Result **17.46°**

code: collector_design()

C65 Size of one collector

Formula $A_1 = A / N$; side = $\sqrt{A_1}$

Values $2.11 \times 10^{22} / 5.00 \times 10^{6}$

Result **4.22×10^{15} m², about 64,950 km × 64,950 km**

code: collector_design()

C66 Mass of the belt and of one collector

Formula $m = A \cdot \sigma_{areal}$

Values $2.11 \times 10^{22} \times 0.1$ kg/m²

Result **2.11×10^{21} kg (0.0287 Moon masses); one collector 4.22×10^{14} kg**

code: collector_design()

C67 Sunlight at the belt

Formula $S = S_{1AU} / r^2$

Values $1361.0 / 0.5^2$

Result **5,444.0 W/m²**

code: collector_operating()

C68 **Sunlight intercepted (whole belt built)**

Formula $L_{cap} = f \cdot L_{Sun}$

Values $0.3 \times 3.828 \times 10^{26}$

Result **1.15×10^{26} W**

code: collector_power()

C69 **Panel temperature (energy balance, both faces radiate)**

Formula $T = [(a\,S - \eta(T)\,S) / (2\,\varepsilon\,\sigma)]^{1/4}$, $\eta(T) = \eta_{25°C}(1 - k\,(T - 298.15\ K))$

Values $a = 0.9$, $\varepsilon = 0.85$, $\eta_{25°C} = 0.25$, $k = 0.002/K$ (solved by iteration)

Result **T = 450.0 K, η(T) = 17.41 % (×0.696); idle panel (no electricity) 474.8 K**

code: collector_operating()

C70 **Ageing of the fleet (panels replaced at end of life)**

Formula $f_{age} = 1 - d \cdot L_{life} / 2$

Values $1 - 0.002 \times 50 / 2$

Result **0.950**

code: collector_operating()

C71 **Operating solar-cell efficiency**

Formula $\eta_{op} = \eta(T) \cdot f_{age}$

Values 0.1741×0.950

Result **16.54 %**

code: collector_operating()

C72 **Panel electricity (whole belt, clear lanes subtracted)**

Formula $P_{pv} = f\,L_{Sun}\,(1 - lanes)\,\eta_{op}$

Values $1.15 \times 10^{26} \times (1 - 5.11 \times 10^{-5}) \times 0.1654$

Result **1.90×10^{25} W; one collector 3.80×10^{18} W**

code: collector_power()

C73 **Light-pressure ratio of a collector (push of sunlight / pull of the Sun)**

Formula $\beta = L_{Sun} / (4\pi\,c\,GM_{Sun}\,\sigma_{areal})$

Values $3.828 \times 10^{26} / (4\pi \times 2.998 \times 10^{8} \times 1.327 \times 10^{20} \times 0.1)$

Result **0.00766 (0.77 %)**

code: collector_design()

C74 **Orbital period of a collector**

Formula $P = r^{3/2} / \sqrt{(1 - \beta)}$ years

Values $0.5^{1.5} / \sqrt{(1 - 0.0077)}$

Result **0.3549 years = 129.6 days**

code: collector_design()

C75 **Gravity of the whole belt on Earth (vs the Sun's pull)**

Formula $a = \Sigma\,G\,m_i\,d_i/|d_i|^3$ over 6,480 belt points

Result **1.28×10^{-9} of the Sun's pull at 1 AU, 1.09×10^{-9} at 2.7 AU (negligible)**

code: belt_pull_on_earth()

Instrument E2.2 - Microwave phased-array transmitters

Use: each collector turns its electricity into a 5.8 GHz beam with a 30 km phased array and points it at the arms. Efficiency 80 %; the beams are throttled to what the launcher needs.

C76 Wavelength

Formula $\lambda = c / f$

Values $2.99792\times10^{8} / 5.80\times10^{9}$

Result **5.169 cm**

code: beam_capture_efficiency()

C77 Microwave power sent (whole belt, full output)

Formula $P_{mw} = P_{pv}\,\eta_{mw}$

Values $1.90\times10^{25} \times 0.8$

Result **1.52×10^{25} W; per transmitter 3.04×10^{18} W**

code: collector_power()

C78 Intensity on a transmitter aperture at full output (indicative)

Formula $I = (P_{mw}/N) / (\pi\,(D_t/2)^2)$

Values $3.04\times10^{18} / 7.07\times10^{8}$ m²

Result **4.30×10^{9} W/m² = 3.16×10^{6} × sunlight at Earth**

code: indicative

C79 Beam capture at 0.50 AU (Goubau relation)

Formula $\tau = D_t\,D_r / (\lambda\,d);\ \eta_{cap} = 1 - e^{-\tau^2}$

Values $\tau = 30{,}000 \times 1.00\times10^{6} / (0.05169 \times 7.480\times10^{10}) = 7.759$

Result **η_{cap} = 100.00 %**

code: beam_capture_efficiency()

C80 Central beam spot diameter at 0.50 AU (indicative, Airy)

Formula $D_{spot} \approx 2.44\,\lambda\,d / D_t$

Values $2.44 \times 0.05169 \times 7.480\times10^{10} / 30{,}000$

Result **314 km (rectenna 1000 km)**

code: indicative

C81 Beam capture at 1.50 AU (Goubau relation)

Formula $\tau = D_t\,D_r / (\lambda\,d);\ \eta_{cap} = 1 - e^{-\tau^2}$

Values $\tau = 30{,}000 \times 1.00\times10^{6} / (0.05169 \times 2.244\times10^{11}) = 2.587$

Result **η_{cap} = 99.88 %**

code: beam_capture_efficiency()

C82 Central beam spot diameter at 1.50 AU (indicative, Airy)

Formula $D_{spot} \approx 2.44\,\lambda\,d / D_t$

Values $2.44 \times 0.05169 \times 2.244\times10^{11} / 30{,}000$

Result **943 km (rectenna 1000 km)**

code: indicative

C83 **Average capture over the whole belt, Earth at 1 AU / at the target**

Formula $\bar{\eta}_{cap}$ = mean over 6,480 belt points (beams within 2 R_{Sun} of the Sun blocked)

Values blocked: 0.1 % / 0.1 % of the belt

Result **99.88 % / 86.30 %**

code: collector_power()

C84 **Pointing accuracy needed (indicative)**

Formula $\delta\theta \approx (D_r/2) / d_{max}$

Values 5.00×10^{5} m / 4.787×10^{11} m

Result **1.04×10^{-6} rad**

code: indicative

C85 **Tested mission: transmitter waste heat and beam spill**

Formula $Q_{tx} = E_{el}(1-\eta_{mw})$; $E_{spill} = E_{mw}(1-\eta_{cap})$

Values 2.65×10^{35} × 0.20; 2.12×10^{35} − 2.07×10^{35}

Result **$Q_{tx} = 5.30\times10^{34}$ J; spill 5.38×10^{33} J (2.54 %), of which 1.00×10^{-9} reaches Earth**

code: settle_interval()

Instrument E2.3 - Rectennas on the 8 arm tips

Use: receive the microwaves (only the arms receive power) and turn them back into electricity with 85 % efficiency. Diameter 1000 km each.

C86 **Area of one rectenna**

Formula $A_r = \pi (D_r/2)^2$

Values $\pi \times (5.00\times10^{5})^2$

Result **7.85×10^{11} m² (0.0493 × Earth's disc for all 8)**

code: collector_design()

C87 **Microwaves arriving (full output), Earth at 1 AU / at the target**

Formula $P_{in} = P_{mw}\,\bar{\eta}_{cap}$

Values 1.52×10^{25} × 0.9988; 1.52×10^{25} × 0.8630

Result **1.52×10^{25} W / 1.31×10^{25} W**

code: collector_power()

C88 **Electricity delivered to the arms (full output)**

Formula $P_{elec} = P_{in}\,\eta_r$

Values 1.52×10^{25} × 0.85

Result **1.29×10^{25} W at 1 AU, 1.11×10^{25} W at 2.7 AU; at year 0 (only 30 % built) 3.87×10^{24} W**

code: collector_power()

C89 **Supply vs the launcher's peak demand**

Formula $P_{elec,min} / P_{peak}$

Values 1.11×10^{25} / 6.34×10^{24}

Result **1.76 ×**

code: feasibility()

C90 **Beam intensity on a rectenna at full output (Earth at 1 AU)**

Formula $I = P_{in} / (8\ A_r)$

Values 1.52×10^{25} / ($8 \times 7.85\times10^{11}$)

Result **2.42×10^{12} W/m² = 1.77×10^{9} × sunlight**

code: collector_design()

C91 **Tested mission: average and peak beam power received**

Formula $\bar{P} = E_{captured} / t$

Values 2.07×10^{35} J / (3,600 y)

Result **mean 1.82×10^{24} W (2.27×10^{23} W per arm, 2.89×10^{11} W/m²); peak 1.83×10^{24} W**

code: summary() beam_mean_W

C92 **Tested mission: rectenna heat**

Formula $Q_r = E_{captured}\ (1 - \eta_r)$

Values $2.07\times10^{35} \times 0.15$

Result **3.10×10^{34} J (removed by the E1 radiators)**

code: settle_interval()

C93 **Tested mission: push of the absorbed microwaves on Earth (indicative side effect)**

Formula $\Delta v = E_{in} / (c\ M)$

Result **114 m/s (counted in the report, not applied)**

code: summary() beam_push_dv_ms

Instrument E2.4 - Sail-tilt station keeping

Use: each collector tilts slightly so the sunlight push corrects its orbit (no propellant). It may use up to collector_sk_authority = 0.5 of its light push. 12 collectors are simulated with it and the same 12 without it (RK4, half-day steps) along Earth's and Mars's simulated motion.

C94 **Sun's pull at the belt**

Formula $g = GM_{Sun} / R^2$

Values 1.3271×10^{20} / $(7.480\times10^{10})^2$

Result **0.0237 m/s²**

code: BeltKeeper

C95 **Largest control acceleration**

Formula $a_{ctrl} = k_{sk}\ \beta\ GM_{Sun} / R^2$

Values $0.5 \times 0.00766 \times 0.0237$

Result **9.08×10^{-5} m/s² = 0.00383 of the Sun's pull**

code: BeltKeeper.auth

C96 **Controller speed (PD gains)**

Formula $\omega_c = 4\ n,\ n = \sqrt{(GM_{Sun}(1-\beta)/R^3)}$

Values $n = 5.61\times10^{-7}$ rad/s

Result **$\omega_c = 2.24\times10^{-6}$ rad/s (time constant 5.2 days)**

code: BeltKeeper.wc

C97 **Beam recoil on a collector (full output)**

Formula $a_{rec} = (P_{mw}/N) / (c\ m_1)$

Values $3.04\times10^{18} / (2.998\times10^{8} \times 4.22\times10^{14})$

Result **2.40×10^{-5} m/s² = 0.00101 of the Sun's pull**

code: BeltKeeper.advance() a_rec

C98 **Pull of a neighbouring collector**

Formula $a = G\ m_1 / s^2$

Values $6.6743\times10^{-11} \times 4.22\times10^{14} / (6.495\times10^{7})^2$

Result **2.81×10^{-10} of the Sun's pull**

code: BeltKeeper.neighbour_ratio

Tested mission (3,600 years)	WITH station keeping	WITHOUT
largest distance from the slot	730 km	1.54×10^{8} km
... in collector spacings (64,950 km)	0.0112	2,371
orbit size change |Δa|	1,158 km	25,044 km
largest eccentricity	1.07×10^{-5}	0.109
control effort	96.8 m/s per year	0

Largest disturbances (fraction of the Sun's pull): beam recoil 1.39×10^{-4}, planets 2.48×10^{-6}, neighbours 2.81×10^{-10}; the controller can correct up to 0.00383.

Instrument E2.5 - Clear sunlight lane

Use: the collectors leave a moving gap on the line Sun-Earth (and Sun-Mars) so that Earth sees the whole solar disc.

C99 **Share of the belt that must stay empty for Earth**

Formula $\text{lane} = \rho^2 / (4\ R^2\ f),\ \rho = R_{Sun}(r-R)/r + R_{Earth}\ R/r$

Values r = 1 AU

Result **1.84×10^{-5} (Mars: 3.27×10^{-5})**

code: lane_fraction()

C100 **Power given up for the lanes**

Formula $\Delta P = L_{cap} \times \text{lanes} \times \eta_{op}$

Values $1.15\times10^{26} \times 5.11\times10^{-5} \times 0.1654$

Result **9.70×10^{20} W of electricity**

code: collector_power()

Instrument E2.6 - Collector factories (Mercury material)

Use: 30 % of the belt exists at year 0; factories finish it in 400 years and replace panels after 50 years (95 % recycled). Manufacturing energy comes from the collectors' own output.

C101 **Built share over time**

Formula $b(t) = b_0 + (1 - b_0)\ t / T_{build}\ (\max 1)$

Values $b_0 = 0.3$, T_{build} = 400 y

Result **b(100 y) = 47.5 %, b(200 y) = 65.0 %, complete at 400 y**

code: collector_built()

C102 **Replacement mass per year (whole belt)**

Formula $\dot{m} = m_{belt} / L_{life}$

Values $2.11 \times 10^{21} / 50$

Result **4.22×10^{19} kg per year**

code: collector_design()

C103 **Fresh material over the mission**

Formula $m_{fresh} = m_{belt} + \dot{m}\,(1 - recycle)\,T$

Values $2.11 \times 10^{21} + 4.22 \times 10^{19} \times 0.05 \times 10{,}000$

Result **2.32×10^{22} kg = 7.0 % of Mercury's mass**

code: collector_design()

C104 **Manufacturing energy over the mission**

Formula $E = e_{emb}\,(m_{belt}(1 - b_0) + \dot{m}\,T)$

Values $e_{emb} = 1.00 \times 10^{8}$ J/kg

Result **4.23×10^{31} J = 0.0706 years of the belt's full electricity**

code: collector_design()

Tested mission: manufacturing energy charged 1.47×10^{31} J; panels built 1.47×10^{23} kg, fresh material 8.77×10^{21} kg.

6. Enabler E3 - Space mirror ring (soletta) and the climate

Space mirror ring around Earth (soletta)
1.6e15 m^2 of aluminium-film mirrors (8e12 kg) in a terminator orbit 20 Earth radii out reflect extra sunlight onto Earth: up to 1.10 x today's sunlight at 2.7 AU. The agent sets how much (mirrors not needed are turned so their light passes Earth).

Gap to reality: Physically possible (only reflection, no new physics, no energy needed). Demonstrated small: Znamya-2 (1993) 20 m mirror lit a 5 km spot on Earth; solar sails IKAROS (2010, ~200 m^2). Needed: ~1e13 x that area; film made from asteroid aluminium; formation flying and pointing of a million 40 km mirrors.

Modelled: Capacity (1/r^2), light sent, light turned away, mirror heat (energy ledger), mass; the climate effect. Not the mirrors' orbits, light pressure, pointing errors or the light pattern on Earth.

Instrument E3.1 - Mirror film panels

Use: reflect sunlight onto Earth. Aluminium film, 5 g/m², reflectivity 90 %. Mirrors that are not needed are turned so their light passes Earth.

C105 **Mirror mass**

Formula $m = A\ \sigma_{areal}$

Values $1.60\times10^{15} \times 0.005$

Result **8.00×10^{12} kg**

code: mirror_design()

C106 **Area compared with Earth's disc; equivalent single disc**

Formula $A / (\pi\ R_{Earth}^2)$; $r = \sqrt{(A/\pi)}$

Values $1.60\times10^{15} / 1.28\times10^{14}$

Result **12.55 ×; one disc of radius 22,568 km, or 1,273,240 mirrors of 40 km**

code: mirror_design()

C107 **Area facing the Sun (45° tilt)**

Formula $A_{eff} = A \cos 45°$

Values $1.60\times10^{15} \times 0.7071$

Result **1.13×10^{15} m²**

code: mirror_design()

C108 **Largest light the mirrors can send to Earth, Earth at 1 AU**

Formula $P_{cap} = A_{eff}\ \rho\ S_{1AU} / r^2$

Values $1.13\times10^{15} \times 0.9 \times 1361.0 / 1^2$

Result **1.39×10^{18} W = 7.985 × the sunlight Earth intercepts (1.74×10^{17} W)**

code: mirror_design()

C109 **... Earth at the target (2.7 AU)**

Formula $P_{cap} = A_{eff}\ \rho\ S_{1AU} / a_t^2$

Values $1.13\times10^{15} \times 0.9 \times 1361.0 / 2.7^2$

Result **1.90×10^{17} W = 1.095 × today's sunlight on Earth**

code: mirror_design()

C110 **Mirror light needed at 2.7 AU to restore today's sunlight**

Formula $P_{need} = (1 - 1/a_t^2)\, S_{1AU}\, \pi\, R_{Earth}^2$

Values $(1 - 0.1372) \times 1.74\times10^{17}$

Result **1.50×10^{17} W = 0.788 of the mirror capacity (margin 26.9 %)**

code: climate_integrate()

C111 **Heat absorbed by the mirrors**

Formula $Q = (1 - \rho) \times$ sunlight on the mirrors

Values $1 - 0.9$

Result **10 % of the sunlight they catch (radiated to space)**

code: settle_interval()

C112 **Light pressure on a mirror at 2.7 AU (indicative, NOT simulated)**

Formula $a = 2\, S \cos^2 45° / (c\, \sigma_{areal})$

Values $2 \times 186.7 \times 0.5 / (2.998\times10^8 \times 0.005)$

Result **1.25×10^{-4} m/s² = 0.51 % of Earth's gravity at the mirror orbit**

Meaning Small but not negligible: the mirrors need active orbit and attitude control (part of E3, not simulated).

code: indicative

Instrument E3.2 - Terminator orbit and pointing

Use: the ring orbits Earth at 20 Earth radii in a plane at right angles to the sunlight, so each mirror always sees the Sun at 90° from Earth and is tilted 45°. The agent's action a4 sets how much light is aimed at Earth.

C113 **Orbit radius and period**

Formula $r = k\, R_{Earth}$; $P = 2\pi \sqrt{(r^3/(G\, M_{Earth}))}$

Values $r = 20 \times 6.371\times10^6 = 1.274\times10^8$ m

Result **P = 5.24 days**

code: mirror_design()

C114 **Size of the light spot on Earth from one mirror (Sun's angular size)**

Formula $D = \theta_{Sun}\, r$, $\theta_{Sun} = 2\, R_{Sun} / (a\ AU)$

Values θ_{Sun} at 2.7 AU = 0.00344 rad

Result **439 km at 2.7 AU (1,185 km at 1 AU)**

code: mirror_design()

C115 **Pointing accuracy to keep the spot on Earth (indicative)**

Formula $\delta\theta \approx R_{Earth} / r$

Values $6.371\times10^6 / 1.274\times10^8$

Result **0.0500 rad = 2.86° (a mirror turns by half that)**

code: indicative

Climate model (what the mirrors must achieve)

Two-box energy balance: a surface layer (atmosphere + ocean mixed layer) exchanging heat with the deep ocean; Earth's reflectivity rises from 0.30 to 0.60 as it freezes (snowball risk).

C116 Effective emissivity (fixed greenhouse, 288 K today)

Formula $\varepsilon_{eff} = S_{1AU}(1-\alpha_{warm})/4 / (\sigma$ 288&sup4;)

Values 1361.0 × 0.70 / 4 / (5.670×10^{-8} × 288&sup4;)

Result **0.6105**

code: CLIM_EPS

C117 Surface temperature equation

Formula $C_s\, dT_s/dt = S(1-\alpha(T_s))/4 + q_w/(4\pi R^2) - \varepsilon\sigma T_s$&sup4; $- \gamma(T_s-T_d)$; $C_d\, dT_d/dt = \gamma(T_s-T_d)$

Values C_s = 8.0 W yr m^{-2}K^{-1}, C_d = 100.0, γ = 0.7 W m^{-2}K^{-1}, step ≤ 0.0625 y

Result **surface response time ≈ $C_s/(4\varepsilon\sigma T^3 + \gamma)$ = 2.0 years**

code: climate_integrate()

C118 Equilibrium at 2.7 AU WITHOUT mirrors

Formula solve $S(1-\alpha(T))/4 = \varepsilon\sigma$T&sup4;, $S = S_{1AU}/a_t^2$

Values S = 186.7 W/m² (0.1372 × today)

Result **152.4 K (-121 °C): snowball (frozen to the equator)**

code: climate_equilibrium()

C119 Equilibrium at 2.7 AU with ALL mirror light on Earth

Formula $S = S_{1AU}(1/a_t^2 + P_{cap}/(S_{1AU}\pi R^2))$

Values 0.1372 + 1.0954 = 1.2325 × today

Result **303.5 K — the agent aims less light (a set-point) to stay at 288 K**

code: climate_equilibrium()

C120 Life indicators at the target temperature

Formula $T(lat) = \bar{T} + \Delta T/3 - \Delta T \sin^2(lat)$, ΔT = 45 K; ice where T < 263 K; growth for 278-313 K

Values $\bar{T}$ = 288 K

Result **equator 303 K, pole 258 K, ice line 70.5°, growing area 74.5 %: temperate, like today**

code: life_indicators()

C121 Tested mission: mirror light

Formula sunlight on mirrors = to Earth + turned past Earth + heat

Values 8.20×10^{28} = 1.04×10^{28} + 6.34×10^{28} + 8.20×10^{27} J

Result **surface 288.4 K (largest deviation 2.6 K, 0 years outside the livable band); without mirrors it would be 157 K**

code: settle_interval(), summary()

7. Energy and mass ledger of the tested mission

Flow (J)	Hand-designed mission, high fidelity
IN: sunlight on the collectors	1.25×10^{37}
IN: reserve used	0
IN: sunlight on the mirrors	8.20×10^{28}
reflected by the panels	1.25×10^{36}
collector heat	1.10×10^{37}
manufacturing (new + replaced panels)	1.47×10^{31}
panel electricity used	2.65×10^{35}
transmitter heat	5.30×10^{34}
microwaves sent	2.12×10^{35}
beam spill (missed the rectennas)	5.38×10^{33}
captured by the rectennas	2.07×10^{35}
rectenna heat	3.10×10^{34}
arm electricity	1.76×10^{35}
launcher draw	1.76×10^{35}
packet kinetic energy	8.78×10^{34}
launcher heat	8.78×10^{34}
mirror light onto Earth	1.04×10^{28}
mirror light turned past Earth	6.34×10^{28}
mirror heat	8.20×10^{27}
waste heat reaching Earth	1.24×10^{26}

C122 **Energy balance**

Formula: IN = reflected + collector heat + manufacturing + transmitter heat + spill + rectenna heat + packet KE + launcher heat + mirror flows

Values: 1.25385×10^{37} vs 1.25385×10^{37} J

Result: **relative residual 1.88×10^{-16} (round-off level: no energy created or lost)**

code: settle_interval() residual_rel

C123 **Efficiency of the beaming chain (panel electricity to arm electricity)**

Formula: $\eta = \eta_{mw}\,\bar{\eta}_{cap}\,\eta_{r}$

Values: 1.76×10^{35} / 2.65×10^{35}

Result: **66.3 %**

code: settle_interval()

C124 **Sunlight-to-packet efficiency**

Formula: $E_{\text{packet KE}}$ / $E_{\text{sun on used panels}}$

Values: 8.78×10^{34} / (2.65×10^{35} / 0.1654)

Result: **5.48 %**

Mechanical conservation (N-body, whole mission)	Relative residual
energy (system vs launches, escapes, impacts)	3.55×10^{-8}
momentum	1.90×10^{-11}
angular momentum	1.47×10^{-13}
mass	1.41×10^{-16}

Mass: 2.86×10^{22} kg ejected in 28,560 packets; 2.86×10^{22} kg left the Solar System; 1.00×10^{18} kg hit the Moon; 0 kg fell back to Earth; 0 kg hit Mars.

8. Tested results and the control system

High-fidelity mission (seed 2001)	No ejection	Hand-designed heuristic	Best PPO model
result	deadline	success	success
final a / e	1.0000 AU / 0.0122	2.6903 AU / 0.0072	2.6854 AU / 0.0145
simulated years	10,000	3,600	7,600
asteroid used	0.0 %	95.2 %	99.5 %
energy drawn	0 J	1.76×10^{35} J	1.73×10^{35} J
surface temperature	288.0 K	288.4 K	284.9 K
closest Mars pass	0.3670 AU	0.0502 AU	0.0470 AU
reward (scoring only, not physics)	-53.3	214.7	194.8

The hand-designed strategy reaches the target; the trained PPO checkpoint also reaches it in this test but it is an early lucky policy (later training collapsed). Neither is claimed to be optimal.

How the agent's five actions become physical commands

C125 **Action decoding (each action a_i in −1..1)**

Formula: $\text{throttle} = (a_0+1)/2;\ \theta = \pi\, a_1;\ \varphi = (\pi/2)\, a_2;\ u = u_{min} + (a_3+1)/2\ (u_{max}-u_{min});\ S_{set} = 1 + 0.5\, a_4$

Values: u_{min} = 20,000 m/s, $u_{max} = 5.00\times10^6$ m/s

Result: **packet mass = throttle × $m_{pk,max}$; direction ($\sin\theta\cos\varphi$, $\cos\theta\cos\varphi$, $\sin\varphi$) in radial/along-track/normal axes; mirror set-point 0.5-1.5 × today's sunlight**

Meaning: The agent never sets Earth's velocity; it only commands packets and mirror aim, and physics does the rest.

code: OctoEarthEnv.decode_action()

C126 **Reward per decision (scoring only)**

Formula: $r = w_p\Delta\Phi - w_e(\text{e excess}) - w_i(\text{i excess}) - w_E E_{used}/E_0 - w_m m_{used}/m_0 - w_t - w_{Mars}\cdot\text{pen} - w_{clim}\cdot\text{pen} +$ terminal terms

Values: w_p=100, w_E=10, w_m=5, w_t=0.02, w_{Mars}=20, w_{clim}=2, success 100, failure 100

Result: **a number for learning only; all physical results are reported separately**

code: OctoEarthEnv.step()

9. Instrument summary: use and required performance

Instrument	Use	Required performance (this configuration)
E1.1 Asteroid reservoir	reaction mass	3.00×10^{22} kg (0.408 Moon masses)
E1.2 Launch ring + 8 arms	throw packets	1.00×10^{18} kg at up to 5,000 km/s, 8/yr, η = 0.5; peak 6.34×10^{24} W
E1.3 Power conditioning + reserve	launch energy	reserve 3.00×10^{35} J; beams first
E1.4 Directional radiators	waste heat away from Earth	~1.84×10^{19} m² at 1,000 K; leak ≤ 1.00×10^{-9}
E2.1 Thin-film collectors	sunlight to electricity	5.00×10^{6} × 64,950 km, 2.11×10^{21} kg, η_{op} 16.5 % at 450 K
E2.2 Microwave transmitters	electricity to beam	30 km arrays, 5.8 GHz, η 80 %, capture 86 %-100 %
E2.3 Rectennas (8)	beam to electricity	1000 km, η 85 %; delivers 1.11×10^{25}-1.29×10^{25} W
E2.4 Sail-tilt station keeping	hold slots	up to 9.08×10^{-5} m/s²; tested 730 km max drift
E2.5 Clear lane	Earth sees the whole Sun	1.84×10^{-5} of the belt empty
E2.6 Factories	build + replace	4.22×10^{19} kg/yr; fresh 7.0 % of Mercury
E3.1 Mirror film	reflect sunlight	1.60×10^{15} m², 8.00×10^{12} kg, ρ = 0.9; capacity 1.10 × at 2.7 AU
E3.2 Terminator orbit + pointing	aim the light	20 R_{Earth}, 5.2 d period, pointing ~0.0500 rad
(not an enabler) Mars safety shield	keep Mars away	safe distance 0.0246 AU
(not an enabler) RL agent / control	decide every 100 y	5 actions, 31 observations

What is NOT modelled (honest list)

- **P5 Arm waste heat (~1e24 W, 6e6 x Earth's sunlight)** (open): Only harmless because of the directional arm radiators of enabler E1 (1e-9 leaks to Earth). With an undirected radiator Earth would boil. *Remaining:* No physical radiator design.
- **P6 Launch damage to Earth** (open): Each 1e18 kg packet leaves at ~2,500 km/s carrying ~3e30 J; shock, atmospheric loss and heating are not modelled. *Remaining:* Assumed as part of enabler E1 (launches do not damage Earth).
- **P7 Other planets** (open): Venus, Mercury (near the 0.5 AU belt) and Jupiter are not simulated; the final orbit's long-term stability with Jupiter is not assessed. *Remaining:* Add them to the N-body model.
- **P8 Day length, seasons, tides** (open): Spin, obliquity and tides are not modelled; the mirrors' illumination pattern on Earth is not modelled. *Remaining:* -

10. What happens without each instrument

Each instrument is necessary for a reason. The calculations below show what would happen without it, or if it performed worse, using the same model functions. This is why each one is part of an enabler.

C127 **Without the asteroid reservoir**

Formula $\Delta v = v_\infty \ln(m_{sys} / (m_{sys} - m_{thrown}))$, $m_{thrown} = 0$

Result **Δv = 0: Earth cannot change its orbit (momentum is conserved; there is nothing to throw)**

code: feasibility()

C128 **Launcher limited to 30 km/s (fastest electromagnetic launchers studied)**

Formula $\Delta v_{cap} = \sqrt{(u^2 - v_{esc}^2)} \ln MR$

Values u = 30 km/s, MR = 1.004962

Result **138 m/s = 1.18 % of the required 11,658 m/s**

code: feasibility(max_ejection_speed_ms=3e4)

C129 **Without the energy reserve (energy_budget_j = 0)**

Formula $E_{avail} = P_{elec,min}\ T$ vs E_{min}

Values $1.11\times10^{25} \times 3.16\times10^{11}$ vs 1.66×10^{35}

Result **still enough: 3.52×10^{36} J available. At year 0 (30 % of the belt built) the arms get 3.87×10^{24} W vs 1.56×10^{24} W at full throttle with the strategy speed**

Meaning In the tested mission the reserve was not touched (all launch energy came from sunlight through E2). The reserve is an assumed store of energy with no source in the model; the mission does not need it.

code: feasibility(energy_budget_j=0)

C130 **Radiators that are not directional (~8 % of the heat reaches Earth)**

Formula $T = [(S(1-\alpha)/4 + q/(4\pi R^2)) / (\varepsilon\sigma)]^{1/4}$

Values $q = 0.08 \times 1.05\times10^{24}$ W

Result **T ≈ 8,296 K: Earth's surface would be destroyed**

code: climate balance (indicative)

C131 **Without the collector belt**

Formula E_{avail} = reserve only

Values 3.00×10^{35} J vs 1.66×10^{35} J

Result **passes only because of the assumed reserve; Earth's own sunlight for the whole mission is only 5.48×10^{28} J = 3.29×10^{-7} of the minimum**

Meaning Without E2 the launch energy has no physical source; the belt is what makes the energy come from sunlight.

code: feasibility(solar_collectors=False)

C132 **Transmitters 10 times smaller (3 km)**

Formula $\eta_{cap} = 1 - e^{-\tau^2}$, $\tau = D_t D_r/(\lambda d)$

Values D_t = 3 km (only the product $D_t D_r$ matters, so this equals a 10 times smaller rectenna)

Result **capture 16.8 % at 1 AU, 2.1 % at 2.7 AU (instead of 99.9 % / 86.3 %); the rest is spilled**

code: collector_power()

C133 Rectennas 10 times smaller (100 km)

Formula: same Goubau relation

Values: D_r = 100 km

Result: **capture 16.8 % at 1 AU, 2.1 % at 2.7 AU: arm electricity 2.72×10^{23} W at the target**

code: collector_power()

C134 Without station keeping (tested, same mission)

Formula: RK4 orbits of 12 collectors without the sail-tilt control

Values: 3,600 simulated years

Result: **largest drift 1.54×10^{8} km = 2,371 collector spacings (with it: 730 km): the belt would scramble, beams and lanes could not be kept**

code: BeltKeeper (check 19)

C135 Without the clear lane

Formula: shade = 1 when |z|/r < f (Earth's line of sight to the Sun crosses the band)

Values: Earth in the ecliptic: |z|/r ≈ 0 < 0.3

Result: **shade = 1: Earth and its mirrors get no direct sunlight; only internal heat (~0.087 W/m²) would keep the surface near 40 K**

code: earth_shading()

C136 Without factories (belt stays at 30 %)

Formula: $P_{elec}(b = b_0)$

Values: b_0 = 0.3

Result: **3.87×10^{24} W at 1 AU, 3.34×10^{24} W at 2.7 AU (still above the tested mean demand 1.82×10^{24} W, but aged panels would not be replaced: after 50 years the panels reach their end of life)**

code: collector_power()

C137 Without mirrors at the target

Formula: equilibrium at $S = S_{1AU}/a_t^2$

Values: 0.1372 × today

Result: **152.4 K: snowball Earth (ice to the equator, no growing area)**

code: climate_equilibrium()

C138 Pointing error larger than Earth's angular size from the ring

Formula: $\delta\theta > R_{Earth}/r$

Values: 0.0500 rad

Result: **the mirror light misses Earth: same as having no mirrors (C137)**

code: indicative

11. Instrument data sheets

One sheet per instrument: what it is for, its inputs, the performance it must deliver in this configuration, the calculations behind it (links to C-numbers), what happens without it and how far it is from today's technology.

E1.1 Asteroid reservoir (docked to Earth) (part of enabler E1)

Use	Supplies all reaction mass; used up packet by packet.
Inputs	`asteroid_mass_kg = 3.00×10^22`
Required performance	3.00×10^{22} kg (0.408 Moon masses); at least 1.41×10^{22} kg needed even at the maximum speed
Calculations	C20, C21, C22, C23, C127
Without it	see C127
Gap to reality	Moving and docking a body of 40 % of the Moon's mass is far beyond any technology.

E1.2 Launch ring with 8 arms (mass driver) (part of enabler E1)

Use	Throws packets at the commanded speed and direction; all 8 arms throw one packet together.
Inputs	`max_packet_mass_kg = 1.00×10^18; max_ejection_speed_ms = 5.00×10^6; min_ejection_speed_ms = 20,000; ejections_per_year = 8.0; launch_efficiency = 0.5`
Required performance	1.00×10^{18} kg packets, 2,480 km/s (strategy) up to 5,000 km/s, every 45.7 days; peak power 6.34×10^{24} W
Calculations	C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C42, C43, C44, C45, C46, C47, C128
Without it	see C128
Gap to reality	No known structure; each launch carries ~10^{30}-10^{31} J.

E1.3 Arm power conditioning + energy reserve (part of enabler E1)

Use	Turns arm electricity into launch energy; beams first, reserve second.
Inputs	`energy_budget_j = 3.00×10^35`
Required performance	handles up to 6.34×10^{24} W (7.92×10^{23} W per arm); reserve 3.00×10^{35} J (not used in the tested mission)
Calculations	C48, C49, C50, C129
Without it	see C129
Gap to reality	Planetary-scale power electronics; the reserve has no physical source in the model.

E1.4 Directional arm radiators (part of enabler E1)

Use	Send launcher and rectenna waste heat to space, away from Earth.
Inputs	`arm_heat_to_earth_frac = 1.00×10^-9`
Required performance	reject 1.05×10^{24} W; ~1.84×10^{19} m² at 1,000 K; leak to Earth ≤ 1.00×10^{-9}
Calculations	C51, C52, C53, C130
Without it	see C130
Gap to reality	No radiator design; the leak fraction 1e-9 is assumed.

E2.1 Thin-film solar collectors (5 million) (part of enabler E2)

Use	Intercept sunlight and make electricity.
Inputs	`collector_fraction = 0.3; collector_orbit_au = 0.5; collector_count = 5.00×10^6; collector_areal_density = 0.1; pv_efficiency = 0.25; collector_lifetime_years = 50.0`
Required performance	2.11×10^{22} m², 2.11×10^{21} kg, 450 K panels, η_{op} 16.5 %, 1.90×10^{25} W of electricity at full output
Calculations	C63, C64, C65, C66, C67, C68, C69, C70, C71, C72, C73, C74, C75, C131
Without it	see C131
Gap to reality	A Dyson-swarm-scale project (~10^{13} × all solar panels ever built).

E2.2 Microwave phased-array transmitters (part of enabler E2)

Use	Turn electricity into focused 5.8 GHz beams aimed at the arms; keep spill off Earth.
Inputs	`microwave_efficiency = 0.8; microwave_freq_ghz = 5.8; transmitter_diameter_km = 30.0; beam_spill_to_earth_frac = 1.00×10^-9`
Required performance	30 km aperture, 3.04×10^{18} W each at full output, capture 86.3 %-99.9 %, pointing ~1.04×10^{-6} rad
Calculations	C76, C77, C78, C79, C80, C81, C82, C83, C84, C85, C132
Without it	see C132
Gap to reality	Record power beaming is ~kW over ~km.

E2.3 Rectennas on the 8 arm tips (part of enabler E2)

Use	Receive the beams and turn them into electricity (only the arms receive power).
Inputs	`rectenna_diameter_km = 1000.0; rectenna_efficiency = 0.85`
Required performance	1000 km each, 85 % efficient, up to 2.42×10^{12} W/m² (1.77×10^{9} × sunlight)
Calculations	C86, C87, C88, C89, C90, C91, C92, C93, C133
Without it	see C133
Gap to reality	Beam intensity ~10^{8}-10^{9} × sunlight; no known receiver survives it.

E2.4 Sail-tilt station keeping (part of enabler E2)

Use	Hold each collector in its slot using part of its light push (no propellant).
Inputs	`collector_sk_authority = 0.5`
Required performance	control up to 9.08×10^{-5} m/s²; tested effort 97 m/s per year, drift ≤ 730 km
Calculations	C94, C95, C96, C97, C98, C134
Without it	see C134
Gap to reality	Solar sails demonstrated at m²-km² scale only.

E2.5 Clear sunlight lane (part of enabler E2)

Use	Keep a moving gap so Earth and Mars see the whole Sun.
Inputs	`earth_lane_clear = True`
Required performance	1.84×10^{-5} of the belt kept empty for Earth, 3.27×10^{-5} for Mars
Calculations	C99, C100, C135
Without it	see C135
Gap to reality	Coordinated motion of millions of collectors.

E2.6 Collector factories (Mercury material) (part of enabler E2)

Use	Build the belt up and replace aged panels.
Inputs	`collector_initial_built = 0.3; collector_build_years = 400.0; collector_recycle_frac = 0.95;` `collector_embodied_energy_jkg = 1.00×10^8`
Required performance	4.22×10^{19} kg of panels per year; fresh material 7.0 % of Mercury
Calculations	C101, C102, C103, C104, C136
Without it	see C136
Gap to reality	Self-replicating industry on Mercury is hypothetical.

E3.1 Mirror film panels (part of enabler E3)

Use	Reflect sunlight onto Earth (or past it when not needed).
Inputs	`mirror_area_m2 = 1.60×10^15; mirror_reflectivity = 0.9; mirror_areal_density = 0.005`
Required performance	1.60×10^{15} m², 8.00×10^{12} kg; must send up to 1.50×10^{17} W at 2.7 AU (capacity 1.90×10^{17} W)
Calculations	C105, C106, C107, C108, C109, C110, C111, C112, C137
Without it	see C137
Gap to reality	Physically possible (plain reflection); demonstrated at 20 m (Znamya-2), needed ~10^{13} × that area.

E3.2 Terminator orbit + pointing (part of enabler E3)

Use	Keep the mirrors in sunlight at 20 Earth radii and aim the light (agent action a4).
Inputs	`mirror_orbit_re = 20.0; mirrors = True`
Required performance	orbit period 5.24 days; pointing better than 0.0500 rad; light-pressure disturbance 1.25×10^{-4} m/s²
Calculations	C113, C114, C115, C138
Without it	see C138
Gap to reality	Formation flying of about a million 40 km mirrors.

Mars safety shield (control software, not an enabler)

Use	Forecasts the Earth-Mars distance and overrides the agent's throw when Earth would come closer than the safe distance.
Calculations	C54, C55, C56, C57, C58, C59, C60, C61, C62
Gap to reality	Ordinary software; needs the Mars measurements of section 12.

RL agent PPO + heuristic baseline (control software, not an enabler)

Use	Chooses throttle, direction, throw speed and mirror light every 100 years from 31 observations.
Calculations	C125, C126
Gap to reality	Ordinary software; the tested numbers are in section 8.

12. Measuring instruments the control system needs (ordinary technology, not enablers)

The simulation gives the agent and the safety shield **perfect** knowledge of the state. A real system needs measurements. The instruments below are existing kinds of technology (no enabler needed); the required accuracies are *indicative*, derived from the mission tolerances (1-10 % of each tolerance). They are not simulated.

C139 Earth's orbit size: required accuracy

Formula $\delta a \leq 1$ % of the tolerance; vis-viva: $\delta a = 2a^2 v\, \delta v / \mu \Rightarrow \delta v = \delta a\, \mu / (2\, a^2\, v)$

Values $\delta a = 0.01 \times 0.05$ AU = 74,799 km; a = 2.7 AU, v = 18,126 m/s

Result **position to ~74,799 km, velocity to ~1.68 m/s**

code: indicative

C140 Earth-Mars distance: required accuracy

Formula $\delta d \leq 1$ % of the safe distance; forecast over the shield horizon

Values $0.01 \times 3.67 \times 10^6$ km; horizon 10 years

Result **relative position to ~36,746 km, relative velocity to ~0.116 m/s**

code: indicative

C141 Global surface temperature: required accuracy

Formula $\delta T \leq 10$ % of the tolerance; $\delta S/S = \delta T / (dT/dS)$

Values 0.1 × 5 K; dT/dS ≈ 72 K per ×today (heuristic_climate)

Result **temperature to ~0.5 K; sunlight (incl. mirror light) on Earth to ~0.69 %**

code: indicative

C142 Collector positions: required accuracy

Formula $\delta x \leq 1$ % of the tested drift with station keeping

Values 0.01 × 730 km

Result **~7 km per collector**

code: indicative

Quantity measured	Used by	Instrument (existing kind of technology)	Required accuracy
Earth's position and velocity (6 observations)	agent + shield	radio ranging/Doppler to spacecraft and stations (Deep Space Network type), planetary radar, star astrometry	C139
Orbit size, eccentricity, inclination	agent + success test	computed from the above (orbit determination)	e and i to ~0.0005 / 0.05°
Moon distance and whether it is bound	agent + failure test	lunar laser ranging (cm accuracy today)	~1 km is enough
Asteroid mass left	agent	counting and weighing packets (mass flow meters on the arms)	~1 % of a packet = 1.00×10^{16} kg
Energy reserve / beam power	agent + ledger	electrical power and energy meters on the rectennas and arms	~0.1 %
Mars position relative to Earth (5 observations)	agent + shield	radio tracking of Mars orbiters/landers, radar ranging	C140
Surface and deep-ocean temperature, sunlight on Earth (4 observations)	agent + climate test	satellite radiometers, ocean floats (Argo type), solar irradiance monitors	C141

Beam pointing at the rectennas	E2 transmitters	pilot beam from each rectenna (retrodirective phased array)	~1.04×10^{-6} rad
Collector positions in the belt	E2 station keeping	radio/optical ranging between neighbouring collectors + beacons	C142
Mirror attitude and aim	E3 pointing	star trackers + sun sensors on each mirror	better than 0.0500 rad

13. Appendix: every parameter

Key	Meaning	Value	Unit	Category
asteroid_mass_kg	Initial asteroid mass	3e+22	kg	Engineering assumption
max_packet_mass_kg	Max packet mass	1e+18	kg	Engineering assumption
max_ejection_speed_ms	Max ejection speed (exhaust velocity)	5000000.0	m/s	Engineering assumption
min_ejection_speed_ms	Min ejection speed	20000.0	m/s	Engineering assumption
energy_budget_j	Propulsion energy budget	3e+35	J	Engineering assumption
launch_efficiency	Launch efficiency	0.5	0-1	Engineering assumption
ejections_per_year	Ejection frequency	8.0	1/yr	Engineering assumption
target_au	Target orbit radius	2.7	AU	Mission requirement
mission_years	Mission duration	10000.0	yr	Mission requirement
decision_interval_years	Decision interval	100.0	yr	Simulation setting
tol_a_au	Tolerance on target radius	0.05	AU	Mission requirement
tol_e	Tolerance on eccentricity	0.05	-	Mission requirement
tol_inc_deg	Tolerance on inclination	5.0	deg	Mission requirement
validate_years	Stability check duration	200.0	yr	Mission requirement
randomize_phase	Randomize starting position	True	on/off	Simulation setting
train_episodes	Training episodes	1000	-	RL hyperparameter
ppo_n_steps	PPO steps per update	1000	-	RL hyperparameter
ppo_batch_size	PPO mini-batch size	200	-	RL hyperparameter
ppo_epochs	PPO epochs per update	10	-	RL hyperparameter
ppo_learning_rate	Learning rate	0.0003	-	RL hyperparameter
ppo_gamma	Discount factor (gamma)	0.995	-	RL hyperparameter
eval_every_episodes	Evaluate every	50	episodes	RL hyperparameter
seed	Random seed	1	-	Simulation setting
w_progress	Reward: progress weight	100.0	points	Reward weight (scoring only)
w_eccentricity	Reward: eccentricity penalty	1.0	points	Reward weight (scoring only)
w_inclination	Reward: inclination penalty	1.0	points	Reward weight (scoring only)

w_energy	Reward: energy-use penalty	10.0	points	Reward weight (scoring only)
w_mass	Reward: mass-use penalty	5.0	points	Reward weight (scoring only)
w_time	Reward: time penalty	0.02	points/decision	Reward weight (scoring only)
w_success	Reward: success bonus	100.0	points	Reward weight (scoring only)
w_terminal_error	Reward: final-miss penalty	50.0	points	Reward weight (scoring only)
w_failure	Reward: failure penalty	100.0	points	Reward weight (scoring only)
w_mars_safety	Reward: Mars-proximity penalty	20.0	points	Reward weight (scoring only)
mars_gravity	Mars gravity on	True	true/false	Planetary safety (Mars)
other_planets	Mercury, Venus, Jupiter on	True	true/false	Planetary safety (Mars)
mars_max_tide_ratio	Max Mars tide on Earth (x Moon's tide)	0.01	-	Planetary safety (Mars)
mars_max_moon_perturbation	Max Mars pull on the Moon's orbit	0.001	-	Planetary safety (Mars)
mars_min_distance_km	Hard minimum Earth-Mars distance	2000000.0	km	Planetary safety (Mars)
mars_shield	Mars safety shield	True	true/false	Planetary safety (Mars)
shield_caution_factor	Shield caution factor	2.0	x safe distance	Planetary safety (Mars)
shield_horizon_years	Shield forecast horizon	10.0	yr	Planetary safety (Mars)
solar_collectors	Solar collectors on	True	true/false	Solar collectors (energy supply)
collector_fraction	Share of the Sun's output collected	0.3	0-1	Solar collectors (energy supply)
collector_orbit_au	Collector belt distance from the Sun	0.5	AU	Solar collectors (energy supply)
collector_count	Number of collectors	5000000.0	-	Solar collectors (energy supply)
collector_areal_density	Collector mass per area	0.1	kg/m^2	Solar collectors (energy supply)
pv_efficiency	Solar-cell efficiency	0.25	0-1	Solar collectors (energy supply)
microwave_efficiency	Electricity-to-microwave efficiency	0.8	0-1	Solar collectors (energy supply)
microwave_freq_ghz	Microwave frequency	5.8	GHz	Solar collectors (energy supply)
transmitter_diameter_km	Transmitter antenna diameter	30.0	km	Solar collectors (energy supply)
rectenna_diameter_km	Rectenna diameter (per arm)	1000.0	km	Solar collectors (energy supply)

rectenna_efficiency	Rectenna efficiency	0.85	0-1	Solar collectors (energy supply)
pv_temp_coeff	Solar-cell temperature coefficient	0.002	1/K	Solar collectors (energy supply)
collector_absorptance	Panel absorptance	0.9	0-1	Solar collectors (energy supply)
collector_emissivity	Panel emissivity	0.85	0-1	Solar collectors (energy supply)
pv_degradation_per_year	Panel degradation	0.002	1/yr	Solar collectors (energy supply)
collector_lifetime_years	Panel lifetime	50.0	yr	Solar collectors (energy supply)
collector_recycle_frac	Recycled share of replaced panels	0.95	0-1	Solar collectors (energy supply)
collector_initial_built	Belt built at year 0	0.3	0-1	Solar collectors (energy supply)
collector_build_years	Build-up time of the belt	400.0	yr	Solar collectors (energy supply)
collector_embodied_energy_jkg	Embodied energy of collectors	100000000.0	J/kg	Solar collectors (energy supply)
earth_lane_clear	Keep Earth's sunlight lane clear	True	true/false	Solar collectors (energy supply)
collector_sk_authority	Station-keeping authority	0.5	0-1	Solar collectors (energy supply)
climate_model	Climate model on	True	true/false	Climate + space mirrors
mirrors	Space mirrors on	True	true/false	Climate + space mirrors
climate_target_K	Target surface temperature	288.0	K	Climate + space mirrors
climate_tol_K	Temperature tolerance	5.0	K	Climate + space mirrors
mirror_area_m2	Total mirror area	16000000000000000.0	m^2	Climate + space mirrors
mirror_reflectivity	Mirror reflectivity	0.9	0-1	Climate + space mirrors
mirror_orbit_re	Mirror orbit radius	20.0	Earth radii	Climate + space mirrors
mirror_areal_density	Mirror film mass per area	0.005	kg/m^2	Climate + space mirrors
arm_heat_to_earth_frac	Arm waste heat reaching Earth	1e-09	0-1	Climate + space mirrors
beam_spill_to_earth_frac	Beam spill reaching Earth	1e-09	0-1	Climate + space mirrors
w_climate	Reward: climate penalty	2.0	points per 100 y	Reward weight (scoring only)
hifi_dt_hours	High-fidelity time step	6.0	h	Simulation setting
max_fragments	Max tracked fragments	48	-	Simulation setting

release_radius_m	Packet hand-over distance	2000000000.0	m	Simulation setting
fragment_escape_au	Fragment escape distance	60.0	AU	Simulation setting

142 calculations. End of report.